李成睿 Chengrui Li

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https://JerrySoybean.github.io

https://scholar.google.com/citations?user=SXR3RXIAAAAJ

https://github.com/JerrySoybean

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Research Interests

- Computational neuroscience, neural latent variable models, statistical machine learning.
- Eye tracking experiments and data analysis.
- Fractional calculus signal processing.

Education

Georgia Institute of Technology (Georgia Tech) 2021.08 – 2026.02
Ph.D. in Computational Science & Engineering *Atlanta, USA*
M.Sc. in Mathematics (awarded in 2023.12)

- Got the offer in 2020.08 but deferred to 2021.08 due to COVID-19.
- GPA: **4.00**/4.00
- Advisor: Prof. Anqi Wu
- Thesis title: *Latent variable models and inferencing methods for understanding neural connectivity and latent dynamics*

University of Tennessee, Knoxville (UTK) 2018.08 – 2019.05
Undergraduate Non-Degree Exchange Program *Knoxville, USA*

- GPA: **4.00**/4.00

Sichuan University (SCU) 2016.09 – 2020.06
B.Eng. in Software Engineering (Computational Biology) *Chengdu, China*
B.Sc. in Biological Sciences

- GPA: **3.94**/4.00; Rank: **1**/28; Wu Yuzhang Honored degree
- Advisor: Prof. Yifei Pu & Prof. Wei Deng
- Thesis title: *The application of fractional order image enhancement in computational neuroscience*

Employment

Emory University 2026.01 – present
Postdoctoral Research Consultant (Remote), Dr. Dieter Jaeger's Lab *Atlanta, USA*

- Mouse voltage imaging data analysis in a decision liking task.
- Methods: Independent component analysis, frequency band, latent disentanglement, etc.
- LLM agents for automated research and result visualization.

Meta Platforms, Inc. 2025.06 – 2026.08
Research Scientist, Neural Interfaces: ML *New York City, USA*
ML Algorithm team, CTRL-Labs, Reality Labs

- Meta's sEMG neural band that controls the AI display glasses.
- Multi-device transfer learning.
- Finger gesture classification, handwriting classification, and other downstream tasks.
- Attention mechanism, multi-modal fusion, unified neural latent representation.
- Offline metrics evaluation, failure modes analysis and improvement.
- Data integration experiment, loss function design, model iteration and release leadership.
- Intern mentorship on a project of label quality investigation and annotation improvement via non-causal models and auxiliary devices.

- LLM agent (Claude) development and workflow automation.

Meta Platforms, Inc. 2024.05 – 2024.08
Research Intern, Neuromotor Interfaces: Computational Modeling *New York City, USA*
EMG Foundations team, CTRL-Labs, Reality Labs

- Meta's sEMG neural band that controls the AI display glasses.
- Handwriting classification neural network model architecture.
- Separable convolution, S4D, state-space model, etc.
- Achieved 60% on-device inference latency reduction.

Honors and Awards

Runner-up Poster Award in the Neuro Next Initiative Launch Event, Georgia Tech. 2023
Outstanding Undergraduate Thesis, SCU. 2020
Outstanding Graduates, SCU. 2020
Finalist (<0.3%) + **Frank Giordano Award** (the only 1 out of 14,000), Mathematical Contest in Modeling (MCM/ICM), Consortium for Mathematics and Its Applications. 2019
"Tang Lixin" Lifetime Scholarship, Tang Lixin Education Development Foundation. 2018
First Grade Scholarship, SCU. 2018
National Scholarship, Ministry of Education of the People's Republic of China. 2017
First Grade College Mathematical Contest, SCU. 2017

Research Publications

Peer Reviewed Conference Proceedings

C11 "Learning When Visual Context Matters for Mouse Behavior Analysis"
Weihan Li, Jingyang Ke, Qinheng Pu, Yule Wang, **Chengrui Li**, and Anqi Wu
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026

C10 "A Hitchhiker's Guide to Poisson Gradient Estimation"
Michael Ibrahim*, Hanqi Zhao*, Eli Zachary Sennesh, Zhi Li, **Chengrui Li**†, and Hadi Vafaii†
(*co-first authorship, †co-senior authorship)
The Forty-third International Conference on Machine Learning (ICML), 2026

C9 "A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dynamics and Connectivity"
Chengrui Li, Yunmiao Wang, Yule Wang, Weihan Li, Dieter Jaeger, and Anqi Wu
The Forty-third International Conference on Machine Learning (ICML), 2026 **[Spotlight: 2.2%]**

C8 "Uncovering Semantic Selectivity of Latent Groups in Higher Visual Cortex with Mutual Information-Guided Diffusion"
Yule Wang, Joseph Yu, **Chengrui Li**, Weihan Li, and Anqi Wu
The Fourteenth International Conference on Learning Representations (ICLR), 2026

C7 "Learning Time-Varying Multi-Region Brain Communications via Scalable Markovian Gaussian Processes"
Weihan Li, Yule Wang, **Chengrui Li**, and Anqi Wu
The Forty-Second International Conference on Machine Learning (ICML), 2025 **[Oral: 1.0%]**

C6 "Exploring Behavior-Relevant and Disentangled Neural Dynamics with Generative Diffusion Models"
Yule Wang, **Chengrui Li**, Weihan Li, and Anqi Wu
The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS), 2024

C5 "A Differentiable Partially Observable Generalized Linear Model with Forward-Backward Message Passing"
Chengrui Li, Weihan Li, Yule Wang, and Anqi Wu
The Forty-first International Conference on Machine Learning (ICML), 2024

C4 "Multi-Region Markovian Gaussian Process: An Efficient Method to Discover Directional Interactions Across Multiple Brain Regions"
Weihan Li, **Chengrui Li**, Yule Wang, and Anqi Wu
The Forty-first International Conference on Machine Learning (ICML), 2024

C3 "Forward χ^2 Divergence Based Variational Importance Sampling"
Chengrui Li, Yule Wang, Weihan Li, and Anqi Wu
The Twelfth International Conference on Learning Representations (ICLR), 2024 **[Spotlight: 5%]**

C2 "One-hot Generalized Linear Model for Switching Brain State Discovery"
Chengrui Li, Soon Ho Kim, Chris Rodgers, Hannah Choi, and Anqi Wu
The Twelfth International Conference on Learning Representations (ICLR), 2024

C1 "Extraction and Recovery of Spatio-Temporal Structure in Latent Dynamics Alignment with Diffusion Model"
Yule Wang, Zijing Wu, **Chengrui Li**, and Anqi Wu
The Thirty-seventh Annual Conference on Neural Information Processing Systems (NeurIPS), 2023 **[Spotlight: 3%]**

Journal Articles

J3 "Feature Reconstruction Guided Fusion Network for Hyperspectral and LiDAR Classification"
Zhi Li, Ke Zheng, Lianru Gao, Nannan Zi, and **Chengrui Li**
IEEE Transactions on Geoscience and Remote Sensing, 2025

J2 "Cross Semantic Heterogeneous Modeling Network for Hyperspectral Image Classification"
Zhi Li, Ke Zheng, Jiaxin Li, **Chengrui Li**, and Lianru Gao
IEEE Transactions on Geoscience and Remote Sensing, 2024

J1 "Inverse Kernel Decomposition"
Chengrui Li and Anqi Wu
Transactions on Machine Learning Research (TMLR), 2024

Abstracts

A2 "Disentangling time-varying neural latent dynamics and circuit-level connectivity with subspace-attention gating"
Chengrui Li, Kai Lu, Yen-Shuo Su, Jingyang Ke, Chris Rodgers, Robert C. Liu, and Anqi Wu
Society for Neuroscience (SfN), 2026

A1 "Similarity of Memory Representations Modulate Saccade Curvatures"
Golnaz Forouzandehfar, **Chengrui Li**, Aaron T. Buss, and A. Caglar Tas
Journal of Vision, 2024

Preprints

P2 "A Revisit of Total Correlation in Disentangled Variational Auto-Encoder with Partial Disentanglement"
Chengrui Li, Yunmiao Wang, Yule Wang, Weihan Li, Dieter Jaeger, and Anqi Wu
arXiv:2502.02279, 2025

P1 "Continuous-time systems for solving 0-1 integer linear programming feasibility problems"
Chengrui Li and Bruce J. MacLennan
arXiv:1905.04612, 2019

Invited Talks and Other Presentations

Invited Talks

School of Life Sciences, Tsinghua University, Beijing, China 2026.04
"Disentangling Structured, Time-Varying Neural Latent Dynamics and Their Circuit-Level Connectivity"

Data-Driven and Large-Scale Modeling in Neuroscience Workshop, Georgia Tech 2025.04
"One-hot Generalized Linear Model for Switching Brain State Discovery (One-hot HMM-GLM GitHub

Repository on Cybershuttle)"

School of Life Sciences, Tsinghua University, Beijing, China 2024.08
"Latent Variable Models and Inferencing Methods for Understanding Neural Functional Connectivity"

Affiliated Mental Health Center, Zhejiang University School of Medicine (Hangzhou Seventh People's Hospital), Hangzhou, China 2023.08
"Latent Variable Models for Neural Spike Train Data"

Conference Presentations

Poster, COSYNE 2026, Lisbon, Portugal 2026.03
"A disentangled low-rank RNN framework for uncovering neural connectivity and dynamics"
"Uncovering semantic selectivity of latent groups in higher visual cortex with diffusion models"
"Learning time-varying multi-region brain communications via scalable Markovian Gaussian processes"

Poster, COSYNE 2025, Montreal, Canada 2025.03
"Disentangling partially independent low-rank neural representations and connectivity"

Poster, COSYNE 2024, Lisbon, Portugal 2024.02
"One-hot Generalized Linear Model for Switching Brain State Discovery"
"Extraction and recovery of spatio-temporal structure in neural alignment via diffusion models"

Poster, Neuro Next Initiative Launch Event, Georgia Tech 2023.10
"One-hot Generalized Linear Model for Switching Brain State Discovery"

Oral, The 11th International Conference on Brain Informatics (BI 2018), Arlington, TX, USA 2018.12
"The Power and Beauty of Mathematics: A Prospect of Nature Inspiration & Computational Model from the Interdisciplinary View"

Miscellaneous Experience

Teaching Experiences

Grader. MATH 6647 Numerical Dynamical System @ Georgia Tech. Spring 2025
Teaching assistant. CSE 6740 Computational Data Analysis @ Georgia Tech. Fall 2023
Teaching assistant. CSE 6740 Computational Data Analysis @ Georgia Tech. Spring 2023
Graduate student research mentor. SURE @ Georgia Tech. Summer 2022

Summer Schools

CNeuro 2020: Theoretical and Computational Neuroscience Summer School, Tsinghua University. 2020.08
The Chinese University of Hong Kong Summer Workshop, Hong Kong SAR, China. 2018.08
Cognitive Neuroscience Summer School, Peking University. 2018.07

Other Research Experiences

Eye tracking experiment: Response Selection 2018.09 – 2019.10
- Designed the eye-tracking program by Experiment Builder. Completed an eye-tracking data analysis program in MATLAB. Used the eye-tracking technique to investigate the influence of different stimulus-response conditions on the eye movement trajectories.

Web project development 2019.07
- Developed a web project for an e-commerce platform under the SSM framework. Java + MySQL + JSP was used for full-stack agile development.

Clinical internship at the State Key Laboratory of Biotherapy, West China Hospital, SCU 2018.02
- Conducted tests including the Mini-Mental State Examination (MMSE), the Montreal Cognitive Assessment (MoCA), the Hamilton Anxiety Rating Scale (HAM-A), and the Hamilton Depression Rating Scale (HAM-D).

Academic Services

Area Chair

- KDD AI4Sciences Track: 2026 (February), 2027 (Cycle 1)

Reviewer

- NeurIPS: 2023–2026
- ICML: 2023–2026
- ICLR: 2024–2026
- AAAI: 2025–2027
- AISTATS: 2025–2026
- TMLR: 2025–2026
- COSYNE: 2026
- Nature Communications: 2026

Skills

- **Proficient**: Python, PyTorch, Claude Code, Codex, Cursor, Git, MATLAB, LaTeX, Mathematica, Adobe Illustrator, Experiment Builder (eye-tracking)
- **Familiar**: C/C++, Java, SQL, Julia, R, Cinema 4D, Adobe Premiere Pro, Adobe Audition, Linux

Hobbies

Violin

New Amsterdam Symphony Orchestra, New York City 2025.10 – present
Georgia Tech Symphony Orchestra 2023–2025

Magic

- Performing since 2009
- Member, International Brotherhood of Magicians (I.B.M.)
- Member, Society of American Magicians

Others

- Piano, Chinese orchestra conducting, YOYO ball, aerial photography, and film/audio post-processing